

Making community dataset into structured dataset

A curious case of Taxonomy and Folksonomy in Gaming Industry

INTRODUCTION

Steam is one of the largest digital game platforms in the world. Its tagging system lets any player apply any label to any game; producing 445 unique community-generated tags across the top 1,000 games. Tags like RPG, Cyberpunk, Co-op, and Pixel Art sit side by side with no hierarchy, no controlled vocabulary, and no dimension labels. This project set out to build a relational database from that data; Publisher, Game, Tag to help make it FAIR and find it more about the taxonomy behind the underlying concept of tags and genres.

"Ask two experts to classify the same game, you get two different answers. Steam's 445 community tags didn't create this problem. They revealed it."

DATASET



Steam 2026 Dataset / top 1,000 Steam games; AppID, Name, Release Date, All Tags, Review Score, Price, Owner Estimates, Peak Players

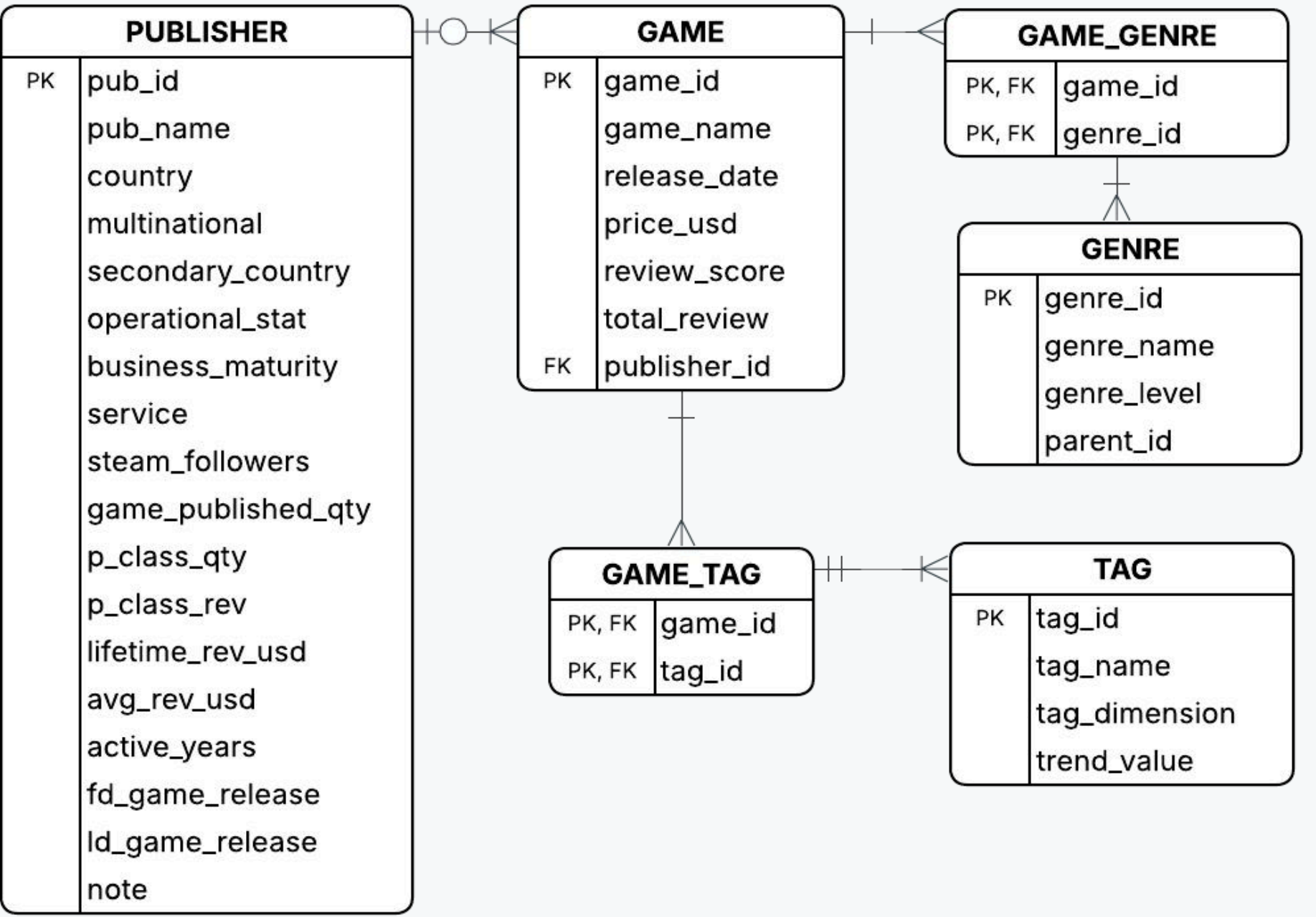


Gamalytics Publisher Data processed into Seyed's Publishers List / 206 Steam publishers; portfolio size, revenue, market-tier classification

METHODOLOGY

- Normalization (1NF → 3NF)**
The raw CSVs stored everything flat. One game row had genres, tags, price, and metadata all crammed together. The ERD broke this apart into atomic tables. This also applied to the publisher.
- Data Type Coercion with Error Handling**
very numeric field (price_usd, review_score, peak_players, 24h) used try/except with None fallback rather than crashing on missing values. This preserves records even when data is incomplete.
- Genre Resolution / Tag Reconciliation**
The CSV used spelling variants the taxonomy didn't recognize ("Co-op" vs. "Co-Op", "Souls-like" vs. "Souls like", "Sci-fi" vs. "Sci-fi"). A normalization map was applied before matching

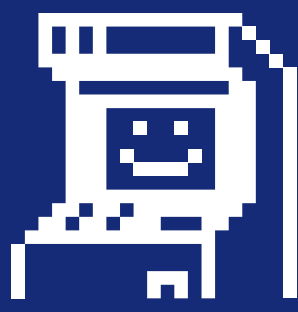
ENTITY RELATIONSHIP DIAGRAM



The TAG table is the core of the taxonomy work. Every Steam tag; whether a genre like RPG or a mood like Relaxing, lives here, classified by dimension and linked upward through TAG_TYPE inspired by SKOS. GENRE sits separately to carry the 3-level hierarchy built from Grelier's framework.

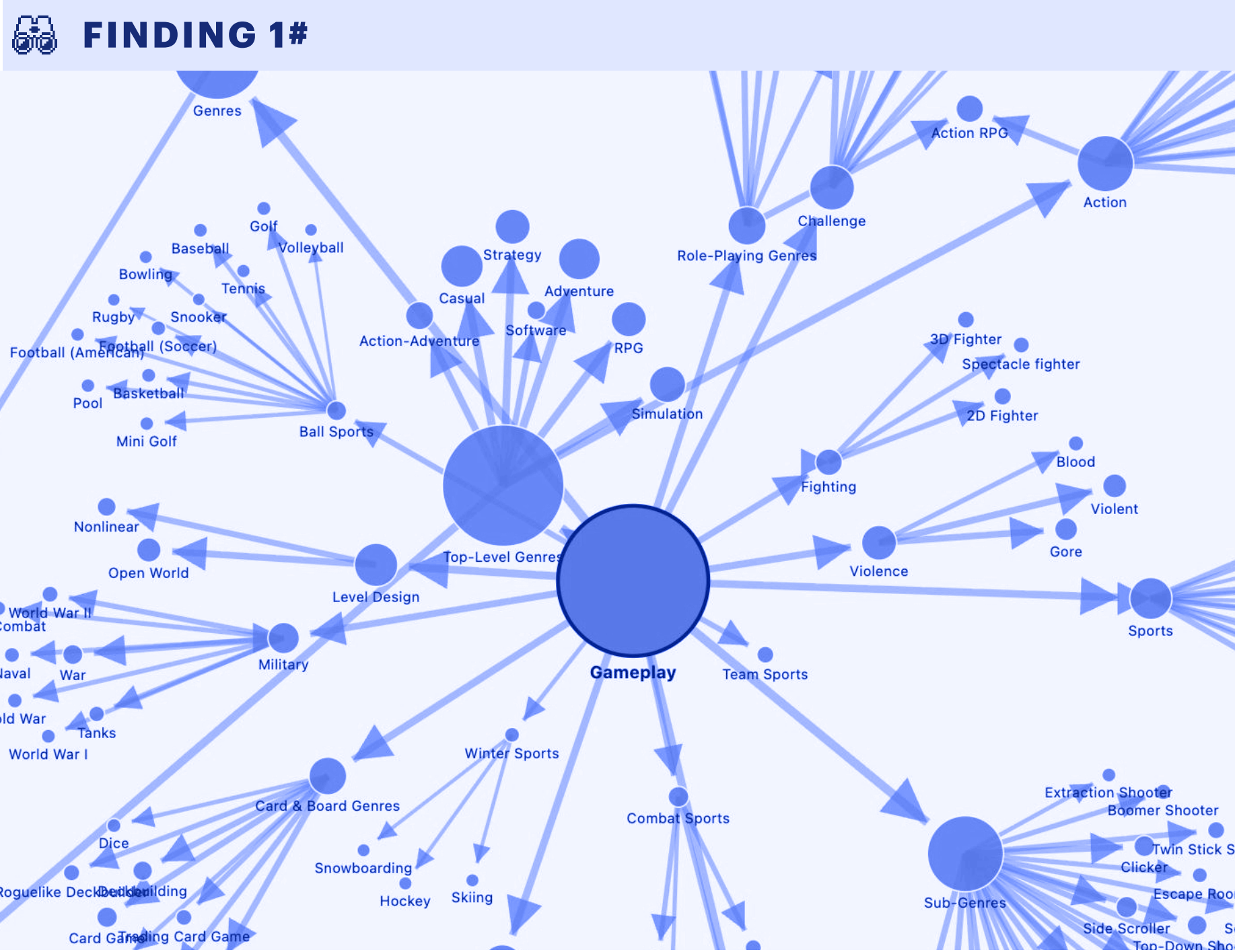
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RESEARCH QUESTION

Can community-curated tags be restructured into a FAIR-compliant for data librarianship?



The Folksonomy as a Network

450 Steam tags mapped by crowd weight. Node size = number of games carrying that tag. No curator decided this structure, but based on the user tags.

- Indie is the single largest Top-Level Genre node** (129,907 games); bigger than Action (105,525), Casual (100,500), and Adventure (95,556) combined
- Themes and Moods is the largest cluster overall** (788,662) — players describe how a game feels more than what it mechanically is.



What Players Tag, and How Often

Each ring = one level of the SteamDB hierarchy. Arc size = cumulative tag weight across Steam's catalog.

- Gameplay → Top-Level Genres → Indie** is the widest arc at the genre ring, confirming that "Indie" functions as the dominant community-assigned genre label
- Visuals and Viewpoint** is the second largest outer arc, players spend more vocabulary describing how a game looks (2D, Pixel Graphics, First-Person)

SOLUTION

Grelier's framework + SKOS to bridge the gap

Grelier, Kaufmann & Schmalz (2023) provided the key insight: Steam tags span 11 non-equivalent dimensions. Treating them all as genres conflates what a game does (mechanic) with what it depicts (theme), how it works (feature), and how it looks (visual).

- Folksonomies CAN be structured:** 445 uncontrolled tags → 132 genres across 9 super-categories, 313 non-genre tags classified rather than discarded
- SKOS (Simple Knowledge Organization System)** bridges both worlds / one schema holds the controlled taxonomy and the flat folksonomy together
- Documentation IS the method** / every edge case logged (Action RPG → RPG; Survival → Theme), satisfying FAIR Reusable principle (Wilkinson et al., 2016)

FINDING 2#

Librarianship dataset use MobyGames as their reference for the genre controlled-vocabularies, while Steam have tags made by the community to facade the game. Two reference systems defined the gap. To understand what the data actually was, two external classification systems were examined side by side. This led me to add another layer of controlled vocabularies from MobyGame and SteamDB.

"A taxonomy is a hierarchical classification of entities, arranged according to presumed natural relationships" — Bates (2002); Mai (2011)



MobyGames / Taxonomy

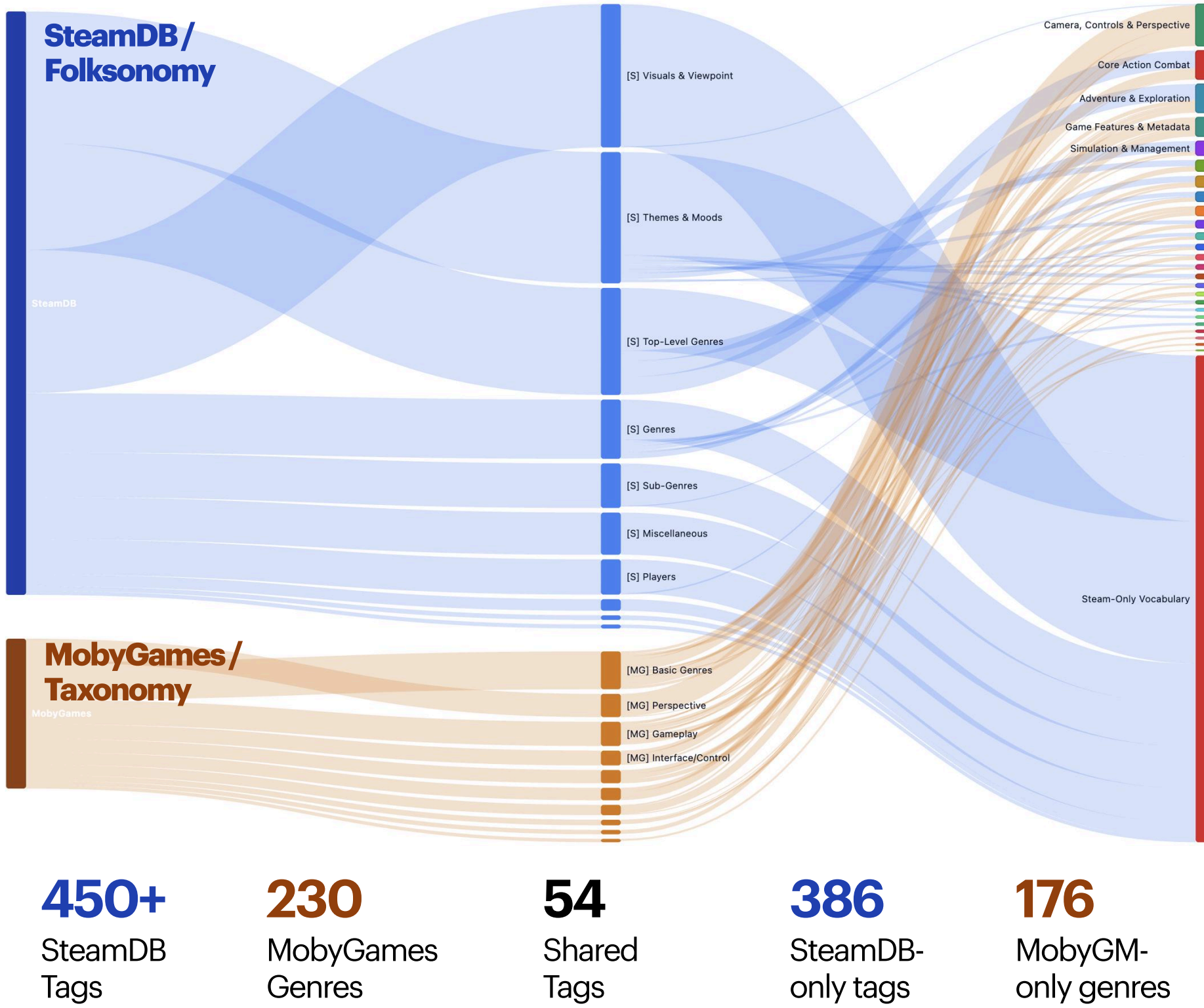
Expert-curated hierarchical classification. Editors assign controlled genre terms from a predefined vocabulary. Top-down, authoritative, stable. Used for validation only.



SteamDB / Folksonomy

User-generated crowd tagging. Tags emerge bottom-up from millions of player votes with no editorial control, no hierarchy, no controlled vocabulary.

DISTRIBUTION OF GENRE & TAGS



CONCLUSION

445 unstructured tags → FAIR-compliant taxonomy
Database: 7 tables, ~10,800 rows

The network structure reflects SteamDB's raw tag-weight distribution across the full Steam catalog, and is not derived from the GENRE or TAG tables in this project's database. The relationship analysis was conducted to help develop a nested relationship for the genre-tag categorization in the ERD.

Community vocabularies and library standards don't naturally get along. Users tag games with language that is expressive and crowd-shaped, "Cozy, Boomer Shooter, Wholesome," terms no cataloguer would invent but that describe real experiences. FAIR principles ask for the opposite: stability, structure, and documented decisions. This project tried to hold both. Steam's 445 tags were not replaced or corrected. They were mapped, classified, and structured without losing what made them useful in the first place. The taxonomy speaks the language of librarians and the language of players at the same time. That balance between how users actually describe things and how collections need to be built is.

ACKNOWLEDGEMENT

Vectors icons are from Streamline, pixel icons. <https://www.streamlinehq.com/>
Set of icons inspired by the work of Susan Kare.

STUDYCASE

STEAM

GENRES Farming Sim Life Sim

RPG Simulation

TAGS Agriculture Crafting

Pixel Graphics Relaxing

Sandbox Multiplayer

MOBY GAMES

GENRE RPG Simulation

GAMEPLAY City building

Construction sim

PERSPECTIVE Diagonal-down

VISUAL 2D scrolling

INTERFACE Direct control

THIS PROJECT

GENRE TABLE Farming Sim RPG

Life Sim Simulation

ATMOSPHERE & TONE Relaxing

SIMULATION Agriculture Craft

PLAYER one player

Multiplayer

VISUAL STYLE Pixel 2D

WORLD Sandbox